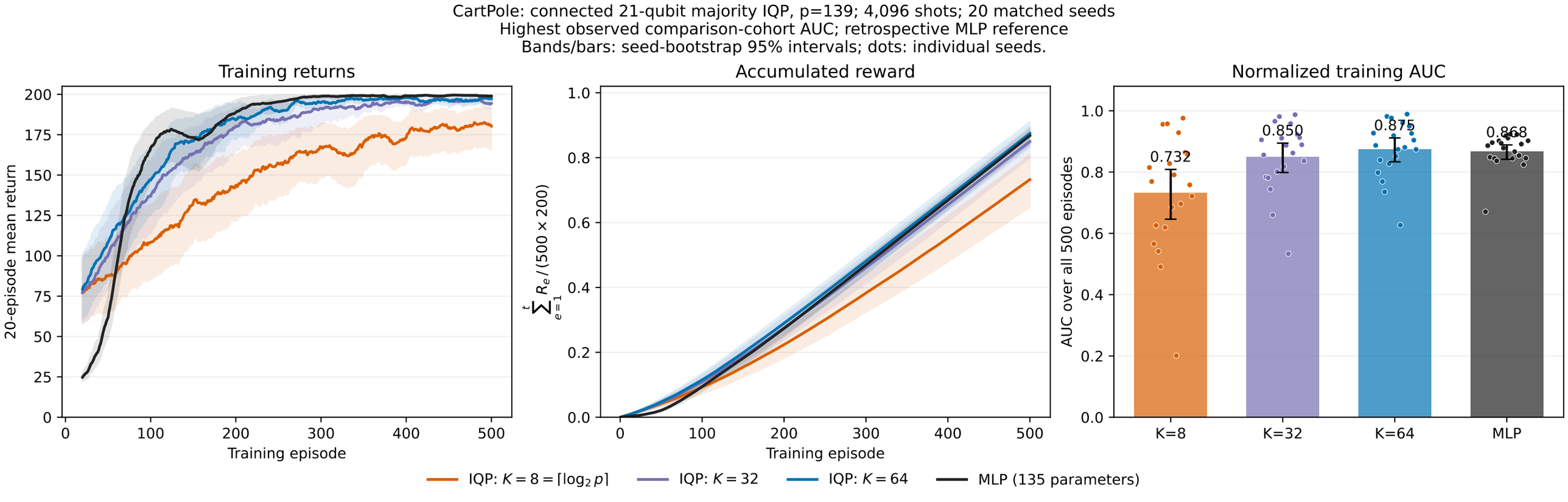

CartPole: connected 21-qubit majority IQP, p=139; 4,096 shots; 20 matched seeds
Highest observed comparison-cohort AUC; retrospective MLP reference
Bands/bars: seed-bootstrap 95% intervals; dots: individual seeds.
Training returns
20-episode mean return
Training episode
Accumulated reward
$\sum_{e=1}^{t} R_e / (500 \times 200)$
Training episode
Normalized training AUC
AUC over all 500 episodes
0.732
0.850
0.875
0.868
K=8
K=32
K=64
MLP
IQP: $K = 8 = \lceil \log_2 p \rceil$
IQP: $K = 32$
IQP: $K = 64$
MLP (135 parameters)